

Clean Log Documentation

Data Version Control:

- Telco_churn_data_v1_raw
- Telco_churn_data_v2_process
- Telco_churn_data_v3_clean

Data Cleaning Process:

Step 1: Initial Inspection & Cleaning

Data cleaning work is processed on the Power Query Editor:

- **Initial rows & columns:** 7043 rows & 38 columns
- **Inspect duplicates: check unique values in the 'Customer ID' column:** 7043 unique values
- **Missing values:**
 - **Numerical columns:** Avg Monthly Long Distance Charges, Avg Monthly GB Download: Replace null values with 0
 - **Categorical columns:** Multiple Lines, Internet Type, Online Security, Online Backup, Device Protection Plan, Premium Tech Support, Streaming TV, Streaming Movies, Streaming Music, Unlimited Data, Churn Category, Churn Reason: Replace missing values with "No Phone Service" in the Multiple Lines column, replace missing values with "No Internet Service" in the internet-related columns, and replace missing values with "No churn" in the Churn related columns.
- **Remove columns:** Zip Code, Longitude, Latitude: for the analysis, focus on 'City' data for geographical information.

- **Data types:** Convert zipcode from integer to text because numerical values can remove leading zeros
- **Standardize column names:** Convert column names by naming convention: transform to lowercase & replace spaces with underscores
- **Applied Clean & Trim to categorical columns** to remove extra spaces and non-printable characters.
- **Handle Null values in the 'Offer' columns:** replace null values with 'No Offer'.
- **Check spelling for categorical columns:** OK
- **Detect outliers in numerical columns:** Age, Number of Dependents, Number of Referrals, Tenure in Months, Avg Monthly Long Distance Charges, Avg Monthly GB Download, Monthly Charge, Total Charges, Total Refunds, Total Extra Data Charges, Total Long Distance Charges, Total Revenue: apply a boxplot to see outliers in a column: leave them as is.

Step 2: Data Logical Validation

Data logical validation work is processed on the Excel & Power Query Editor:

- Customer_id: must be unique values without duplicates
- Total revenue: should be equal to the sum of total charges - total refunds + total long distance charges + total extra data charges: create a new column with a formula and replace inaccurate values
- Age range: should be between 18 and 100
- Tenure in months, total revenue, monthly charge: not negative values; filter them out or delete if they exist
- Total revenue > total charges > monthly charge in a logical sequence: TRUE
- Total charges = monthly charge * tenure in months: create a new column and replace incorrect values (6405), apply a formula = COUNTIF(A:A, FALSE) to see how many incorrect values are in the column

- Zipcode: must have 5 digits based on the US postal code system
- Validate latitude & longitude values if they are in a valid range: latitude valid range: -90 to 90 & longitude valid range: between -180 and 180
- Service dependency validation: internet type populated, but internet service =No, streaming service enabled without internet, and multiple lines enabled without phone service
- Final rows & columns: 6923 rows & 38 columns

Step 3: Final Validation & Save Cleaned Data

- Validate the dataset
- Save the cleaned data for visualization or analysis for insights

Step 4: Data Transformation & Feature Engineering

Create new columns:

- is_churned: Boolean values(0,1) based on customer status
- contract_risk_level: measure potential churn risk based on the contract type
- tenure_band: group values in the tenure_in_months by early, mid, and long-term tenure
- age_group: group age values in the age column by business impact
- customer_lifecycle_stage: group values in tenure_in_months
- service_count: count the services which are provided to customers by the telecom company
- data_usage_tier: group values in the gb_download based on the usage by customers
- early_churn_risk: boolean values (0, 1) measured by customers who are less than 1 year in tenure in months and a month-to-month contract type
- connectivity_type: measured values (1, 2) by customers who are using phone_service or internet_service
- has_premium_service: measured values(0, 1) by customers who are using services

- **streaming_bundle**: count services, how many customers are using streaming services: TV, movie, or music
- **churn_risk_score**: measured value by the contract, tenure_in_months, internet_type, has_premium_service, monthly_charge, number of referrals, streaming_bundle
- **churn_risk_category**: group values by churn_risk_score
- **customer_value_tier**: group values by the calculated values, that is, the monthly_charge multiplied by the tenure_in_months
- **service_tier**: group values by service counts
- **family_status**: group values by people who are married or have dependents
- **engagement_segment**: group values by the engagement_score, which is measured by age, tenure_in_months, and service_counts, normalized and applied a percentage based on the business impact for each
- **Is_refunded**: Boolean values(0,1) which indicate if the customer is refunded or not from the total refunds column

Data Dictionary from Original Data

Table 1: Demographics

- **CustomerID**: A unique ID that identifies each customer.
- **Count**: A value used in reporting/dashboarding to sum up the number of customers in a filtered set.
- **Gender**: The customer's gender: Male, Female
- **Age**: The customer's current age, in years, at the time the fiscal quarter ended.
- **Senior Citizen**: Indicates if the customer is 65 or older: Yes, No
- **Married**: Indicates if the customer is married: Yes, No
- **Dependents**: Indicates if the customer lives with any dependents: Yes, No.
Dependents could be children, parents, grandparents, etc.

- **Number of Dependents:** Indicates the number of dependents that live with the customer.

Table 2: Location

- **CustomerID:** A unique ID that identifies each customer.
- **Count:** A value used in reporting/dashboarding to sum up the number of customers in a filtered set.
- **Country:** The country of the customer's primary residence.
- **State:** The state of the customer's primary residence.
- **City:** The city of the customer's primary residence.
- **Zip Code:** The zip code of the customer's primary residence.
- **Lat Long:** The combined latitude and longitude of the customer's primary residence.
- **Latitude:** The latitude of the customer's primary residence.
- **Longitude:** The longitude of the customer's primary residence.

Table 3: Population

- **ID:** A unique ID that identifies each row.
- **Zip Code:** The zip code of the customer's primary residence.
- **Population:** A current population estimate for the entire Zip Code area.

Table 4: Services

- **CustomerID:** A unique ID that identifies each customer.
- **Count:** A value used in reporting/dashboarding to sum up the number of customers in a filtered set.
- **Quarter:** The fiscal quarter that the data has been derived from (e.g. Q3).
- **Referred a Friend:** Indicates if the customer has ever referred a friend or family member to this company: Yes, No

- **Number of Referrals:** Indicates the number of referrals to date that the customer has made.
- **Tenure in Months:** Indicates the total amount of months that the customer has been with the company by the end of the quarter specified above.
- **Offer:** Identifies the last marketing offer that the customer accepted, if applicable. Values include None, Offer A, Offer B, Offer C, Offer D, and Offer E.
- **Phone Service:** Indicates if the customer subscribes to home phone service with the company: Yes, No
- **Avg Monthly Long Distance Charges:** Indicates the customer's average long distance charges, calculated to the end of the quarter specified above.
- **Multiple Lines:** Indicates if the customer subscribes to multiple telephone lines with the company: Yes, No
- **Internet Service:** Indicates if the customer subscribes to Internet service with the company: No, DSL, Fiber Optic, Cable.
- **Avg Monthly GB Download:** Indicates the customer's average download volume in gigabytes, calculated to the end of the quarter specified above.
- **Online Security:** Indicates if the customer subscribes to an additional online security service provided by the company: Yes, No
- **Online Backup:** Indicates if the customer subscribes to an additional online backup service provided by the company: Yes, No
- **Device Protection Plan:** Indicates if the customer subscribes to an additional device protection plan for their Internet equipment provided by the company: Yes, No
- **Premium Tech Support:** Indicates if the customer subscribes to an additional technical support plan from the company with reduced wait times: Yes, No
- **Streaming TV:** Indicates if the customer uses their Internet service to stream television programming from a third party provider: Yes, No. The company does not charge an additional fee for this service.
- **Streaming Movies:** Indicates if the customer uses their Internet service to stream movies from a third party provider: Yes, No. The company does not charge an additional fee for this service.

- **Streaming Music:** Indicates if the customer uses their Internet service to stream music from a third party provider: Yes, No. The company does not charge an additional fee for this service.
- **Unlimited Data:** Indicates if the customer has paid an additional monthly fee to have unlimited data downloads/uploads: Yes, No
- **Contract:** Indicates the customer's current contract type: Month-to-Month, One Year, Two Year.
- **Paperless Billing:** Indicates if the customer has chosen paperless billing: Yes, No
- **Payment Method:** Indicates how the customer pays their bill: Bank Withdrawal, Credit Card, Mailed Check
- **Monthly Charge:** Indicates the customer's current total monthly charge for all their services from the company.
- **Total Charges:** Indicates the customer's total charges, calculated to the end of the quarter specified above.
- **Total Refunds:** Indicates the customer's total refunds, calculated to the end of the quarter specified above.
- **Total Extra Data Charges:** Indicates the customer's total charges for extra data downloads above those specified in their plan, by the end of the quarter specified above.
- **Total Long Distance Charges:** Indicates the customer's total charges for long distance above those specified in their plan, by the end of the quarter specified above.

Table 5: Status

- **CustomerID:** A unique ID that identifies each customer.
- **Count:** A value used in reporting/dashboarding to sum up the number of customers in a filtered set.
- **Quarter:** The fiscal quarter that the data has been derived from (e.g. Q3).
- **Satisfaction Score:** A customer's overall satisfaction rating of the company from 1 (Very Unsatisfied) to 5 (Very Satisfied).

- **Satisfaction Score Label:** Indicates the text version of the score (1-5) as a text string.
- **Customer Status:** Indicates the status of the customer at the end of the quarter: Churned, Stayed, or Joined
- **Churn Label:** Yes = the customer left the company this quarter. No = the customer remained with the company. Directly related to Churn Value.
- **Churn Value:** 1 = the customer left the company this quarter. 0 = the customer remained with the company. Directly related to Churn Label.
- **Churn Score:** A value from 0-100 that is calculated using the predictive tool IBM SPSS Modeler. The model incorporates multiple factors known to cause churn. The higher the score, the more likely the customer will churn.
- **Churn Score Category:** A calculation that assigns a Churn Score to one of the following categories: 0-10, 11-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80, 81-90, and 91-100
- **CLTV:** Customer Lifetime Value. A predicted CLTV is calculated using corporate formulas and existing data. The higher the value, the more valuable the customer. High value customers should be monitored for churn.
- **CLTV Category:** A calculation that assigns a CLTV value to one of the following categories: 2000-2500, 2501-3000, 3001-3500, 3501-4000, 4001-4500, 4501-5000, 5001-5500, 5501-6000, 6001-6500, and 6501-7000.
- **Churn Category:** A high-level category for the customer's reason for churning: Attitude, Competitor, Dissatisfaction, Other, Price. When they leave the company, all customers are asked about their reasons for leaving. Directly related to Churn Reason.
- **Churn Reason:** A customer's specific reason for leaving the company. Directly related to Churn Category.